WINDOWS RAM ACQUISITION LAB

Belkasoft RAM Capturer | SHA-256 Memory Image Hashing

Educational digital forensics project | Adapted from original Lab 3 coursework

Project overview

This project documents a guided Windows volatile-memory acquisition using **Belkasoft Live RAM Capturer**. The original assignment includes screenshots of tool preparation, a completed memory dump, and a PowerShell SHA-256 hash calculated for a resulting **.mem** file. It demonstrates acquisition and baseline hash documentation; it does not include analysis of the captured memory.

Tool	Belkasoft Live RAM Capturer
Environment	Windows; instructor-provided acquisition files
Output	Memory dump in .mem format
Integrity documentation	SHA-256 hash calculated with PowerShell
Evidence scope	Coursework screenshots; no live incident investigated

01 Prepare the acquisition utility

The lab instructed me to download an instructor-provided Belkasoft RAM Capturer archive, place it in an evidence folder, extract its contents, and launch **RamCapturer.exe** with administrator privileges. The screenshot records the Windows context menu and the **Run as administrator** option.

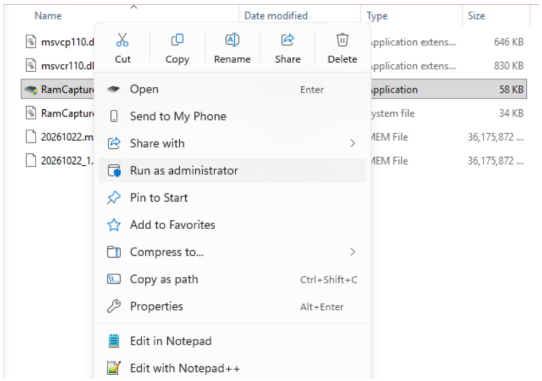


Figure 1. Windows File Explorer showing the extracted RAM Capturer files and elevated launch option.

02 Acquire volatile memory

I used Belkasoft Live RAM Capturer to acquire system memory. The application screenshot shows a full progress bar and the message **“Memory dump completed.”** The tool reports **35,328 MB** of total physical memory dumped. This supports

completion of the acquisition step within the lab.

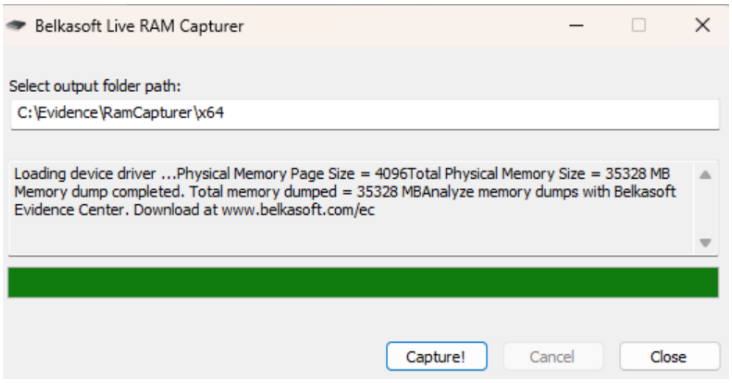


Figure 2. Belkasoft RAM Capturer indicating that the memory dump completed.

03 Calculate the memory image hash

After acquisition, I used the PowerShell **Get-FileHash** cmdlet with the **SHA256** algorithm against a generated **.mem** file. The captured output displays the algorithm, hash value, and file path. Recording a cryptographic digest provides a baseline for later integrity comparisons.

```
PS C:\WINDOWS\System32> Get-FileHash "C:\Evidence\RamCapturer\x64\20261022.mem" -Algorithm SHA256

Algorithm       Hash                                                                   Path
---------       ----                                                                   ----
SHA256          A93C484DA26CE8012B021019AF116AB93E3AAAA1B0EB5DCE760B7C150635214B       C:\Evidence\RamCapturer\x64\2...
```

Figure 3. PowerShell SHA-256 hash calculation for the captured memory image.

04 Interpretation and limitations

The acquisition-complete message and the hash output are two distinct pieces of evidence: the first documents the capture tool reporting completion, while the second documents calculation of a digest for the output file. The coursework does **not** show a second hash calculation or comparison against a previously recorded value, so it should not be described as an independently verified unchanged image.

Skills demonstrated

- Preparing and launching a memory acquisition tool with administrator privileges.
- Capturing volatile Windows memory with Belkasoft Live RAM Capturer.
- Recognizing and documenting successful acquisition status.
- Calculating a SHA-256 digest for a forensic memory image using PowerShell.
- Recording a reproducible workflow with screenshots and identifying the limits of the evidence.

Reflection and next steps

The original lab reflection describes difficulty with some interface steps, followed by successful completion of the acquisition and hashing workflow. A logical next step would be to preserve the recorded SHA-256 digest, recalculate it during evidence transfer, and analyze a sanitized training memory image with an appropriate memory-forensics tool. Those follow-up activities are not claimed as completed in this report.